

# Guernica, éres tu: VR Volumetric Capture as an Art Style

A Recreation of Pablo Picasso's *Guernica* in VR Form

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## ABSTRACT

*Guernica, éres tu* is a first-person anti-war storytelling experience that reconstructs Pablo Picasso's famed painting *Guernica* into a virtual environment. *Guernica, éres tu*, through the use of 8i's volumetric capture technology in a virtual reality environment, the VR application tries to retain the essence of human into the artwork and fits the creative style of abstract and surreal art by modifying the 3D captures. As a result, it persuasively delivers a message that the belligerents, the casualties, and the observers are all actual human beings, not merely statistics nor images on the news.

## CCS CONCEPTS

• **Applied computing** → Arts and humanities; Media arts; • **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Virtual reality.

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## 1 INTRODUCTION

In “*Guernica, éres tu*”, the player follows a set path that introduces the characters of *Guernica*, alongside poetry both handwritten and narrated, with music. The gallery is divided into six sections:

1. The Mother: a mother who has lost her children in the bombing.
2. The Escaper: a civilian surprised by the sudden bombing of the town, trying to escape from it.
3. The Soldier: a soldier with distorted and dismembered limbs sending his final message to his mother
4. The Trapped: a person trapped in building debris, calling for help from his ancestors.
5. The Enlightened: an outsider who brings enlightenment to the scene, but only as an observer.

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Figure 1: Scene “The Trapped” from *Guernica, éres tu*

## 2 THE ESSENCE OF HUMAN AND VOLUMETRIC CAPTURE

While volumetric capture is a costly method of creating models and motion data, it provides a unique feel of humanity due to its capability of accurately capturing very small movements, subtle changes such as breathing, and imperfections of the human body and its movements. This binds especially well with the theme of anti-war, as war is waged between actual human beings: a vital point many willingly forget to alleviate themselves of the psychological cost of fear and depression.

## 3 ABSTRACT ART AND VOLUMETRIC CAPTURE

Volumetric capture allows artists to modify the 3D captures easily which helps when creating surrealist or abstract art. These modifications include scaling, stretching, translating, lighting, clipping, post-processing, and even voxel-based neck model adjustments via third party plugins. Through this technique, implementing unreal effects in a virtual reality scene without the feel of an entirely synthesized or artificial environment and uncanny valleys [Pietroszek and Eckhardt, 2020] is possible.

## 4 VIRTUAL REALITY AND VOLUMETRIC CAPTURE

Augmented reality projects are more likely to use volumetric capture than virtual reality projects, but “*Guernica, éres tu*” proves that VR projects can very well utilize volumetric capture if needed. However, there are some limitations to implementing VC into VR. First is that the file sizes of each capture are very large (ranging from 500MB to 5GB depending on resolution) for around one minute of footage. Second, rendering these models are very costly, and nearly impossible to be run directly on VR devices such as the Oculus Quest 2 without serious frame drops. Thus, volumetric capture requires a PC with decent graphical processing power.

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